Angles, grades and carpentry slope applications

Day 22 • 2026-12-04 • 12:45 PM–2:00 PM

15 questions. Show your method and include units. For multiple choice, enter A, B, C or D. Follow your instructor’s calculator guidance.

Name _____ **Date** _____

Type into the fields and save a copy, or print and write.

01 ANGLE OR GRADE

What is the correct definition of percentage grade?

A. Rise ÷ horizontal run × 100

B. Rise ÷ sloping length × 100

C. Angle in degrees × 100

D. Horizontal run ÷ rise × 100

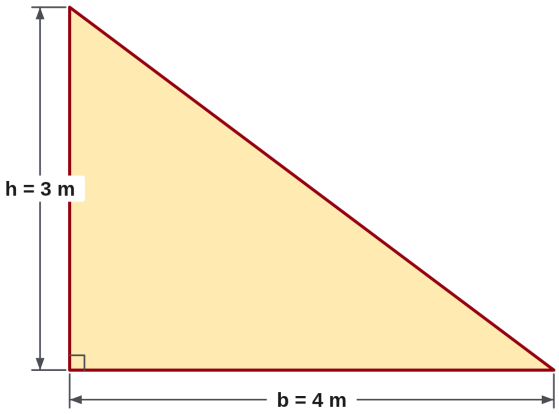
Choice (A–D) _____ **Working / reason** _____

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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02 Using the 3 m rise and 4 m horizontal run in Shape Canvas, calculate the grade.



Shape Canvas

Answer (percent)	Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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03 For an angle calculated from a rise/run ratio, which calculator operation is required?

A. Sine of the length in mm

B. Square root of the angle

C. Inverse tangent in DEG mode

D. Tangent of the percentage



Numeracy Hub

Choice (A–D)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

04 Is a 2% grade equal to a 2-degree angle?

A. Only if lengths use metres

B. No; percent grade and degrees are different measures

C. Yes, always

D. Only for a wooden ramp

Choice (A–D)

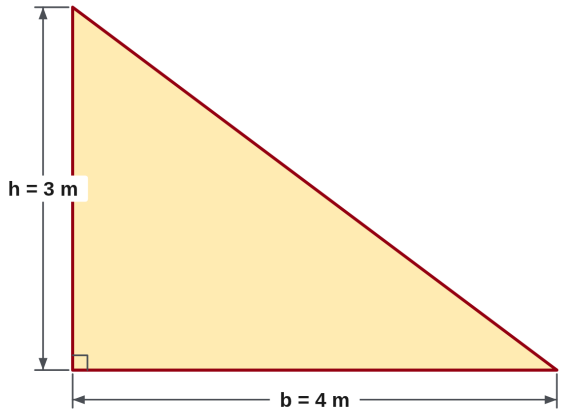
Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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05 Find the angle above the horizontal for the 3 m rise and 4 m run shown. Round to 2 decimal places.



Shape Canvas

Answer (degrees)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

06 CALCULATE SLOPE

A rise is 150 mm over a horizontal run of 3000 mm. Find the grade.

Answer (percent)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

07 A rise is 200 mm over a horizontal run of 4000 mm. Find the angle. Round to 2 decimal places.

Answer (degrees)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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08 A classroom slope has a 2% grade and a 6 m horizontal run. Find its rise in millimetres.

Answer (mm) | **Working / reason**

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

09 A grade is 8% and the vertical rise is 0.24 m. Find the horizontal run.

Answer (m) | **Working / reason**

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

10 Find the angle corresponding to a 10% grade. Round to 2 decimal places.

Answer (degrees) | **Working / reason**

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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11 CHECK THE MEANING

A learner uses a 5 m sloping length as the run when a horizontal run is given. What should change?

A. Use the horizontal run in the grade denominator

B. Use the longest number regardless of meaning

C. Multiply by the hypotenuse

D. Change percent to square metres

Choice (A–D)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

12 A 4% grade over 2500 mm of horizontal run gives what rise?

Answer (mm)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

13 Which pair correctly labels two different slope results?

A. 5 mm grade and 2.86 m angle

B. 5% grade and 5 degrees exactly

C. 5% grade and 2.86-degree angle

D. 5 degrees grade and 2.86% angle

Choice (A–D)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

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14 Can a classroom grade calculation alone establish that a constructed ramp complies with a code?

A. Only the colour of the ramp matters

B. No; applicable requirements must be checked separately

C. Yes, every calculated grade is compliant

D. Yes, if rounded to a whole percent

Choice (A–D)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)

15 An apprentice finds a rise/run ratio of 0.075. What percentage grade should be written?

Answer (percent)

Working / reason

Textbook sections 13.4–13.5 pp.331–333; sections 13.7 pp.337–339 (extension)